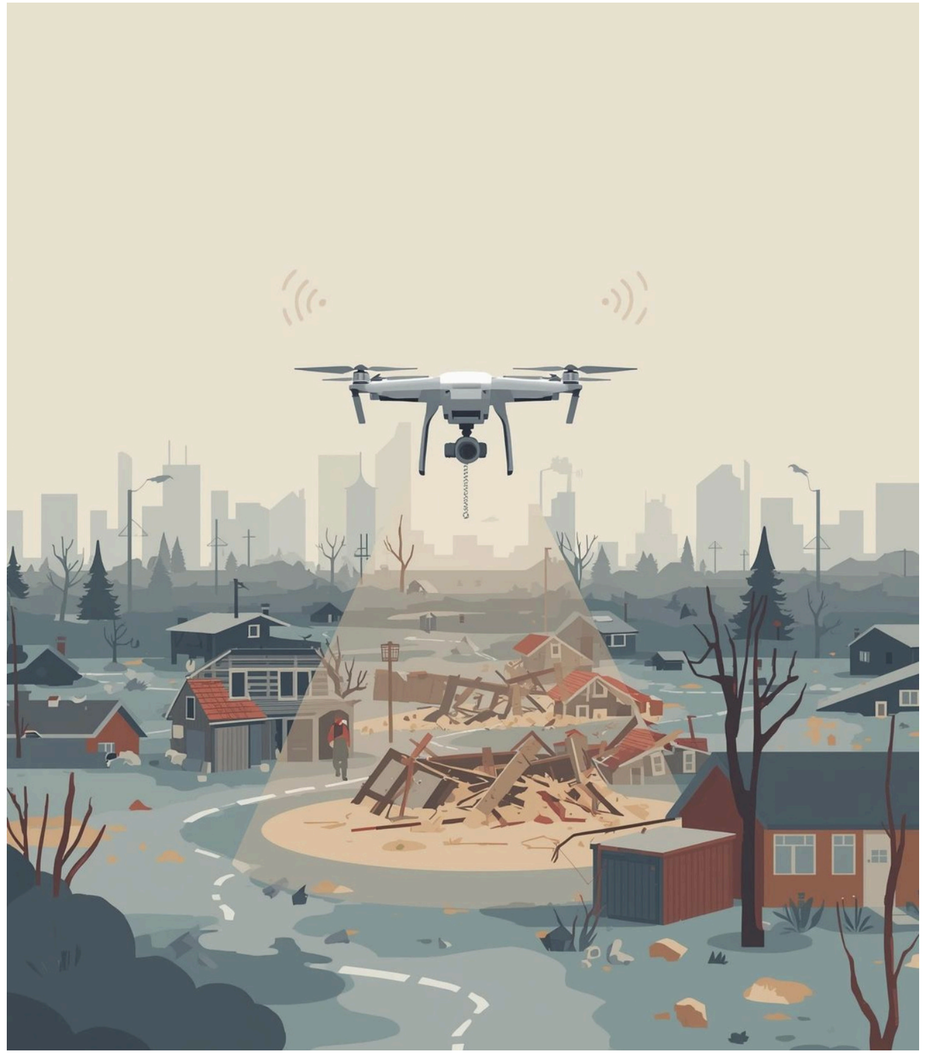
Emergency Communication System During Disaster

SWE 4739 – Embedded Software Development



Team-05



Mst. Iqra Moni
210043109



Takia Farhin
210043117



Labibah Hoque
210043121

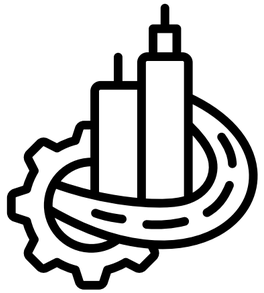


Antara Arifa Mullick
210043132



Tahsin Islam
210043137

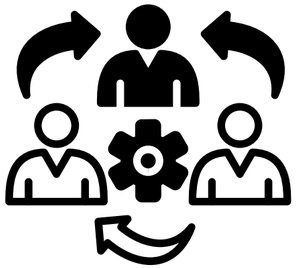
Problem Statement



Infrastructural Damages



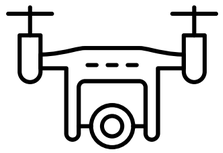
Communication Breakdown



Inefficient Rescue Coordination

Proposed Solution

Utilizing Unmanned Aerial Vehicles (UAV) for Communication



UAV Deployment

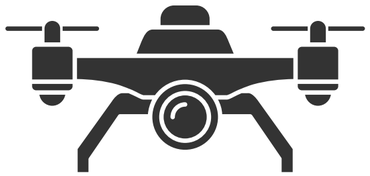
Multiple UAV drones will be deployed to create a **dynamic Flying Ad Hoc Network (FANET)**, ensuring robust communication capabilities in disaster-stricken areas without relying on existing infrastructure.



Communication Bubble

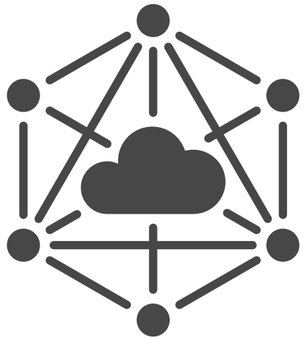
This system will establish a temporary **communication bubble** over affected regions, enabling transmission of SOS signals, GPS coordinates, and images, facilitating efficient coordination for rescue operations.

System Features



Ad Hoc Networking (FANET)

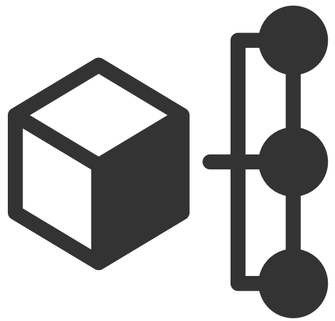
- Multiple small UAVs instead of a single large UAV.
- Self-organizing wireless network.
- Improves survivability and scalability.



Cluster-Based Management

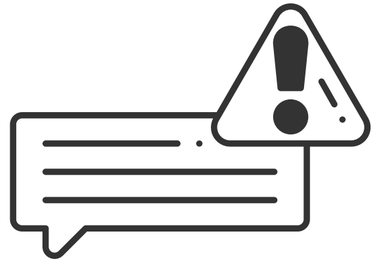
- UAVs grouped into small teams called clusters.
- Reduces wireless traffic congestion.
- Helps manage fast drone movement.

System Features



Dynamic Role Allocation

- Automatic Cluster Head (CH) selection.
- Backup Cluster Head (CH_B) for redundancy.
- Ensures system stability during failures without interruption.



Safety-Critical Communication

- Safety Messages (SMs) prioritized like SOS alerts, survivors location.
- Strict delay constraint: < 100 ms.
- Control commands always transmitted first.

System Features

Adaptive Backoff Mechanism

- Adjusts waiting time before transmission dynamically.
- Based on how many drones are in the cluster.

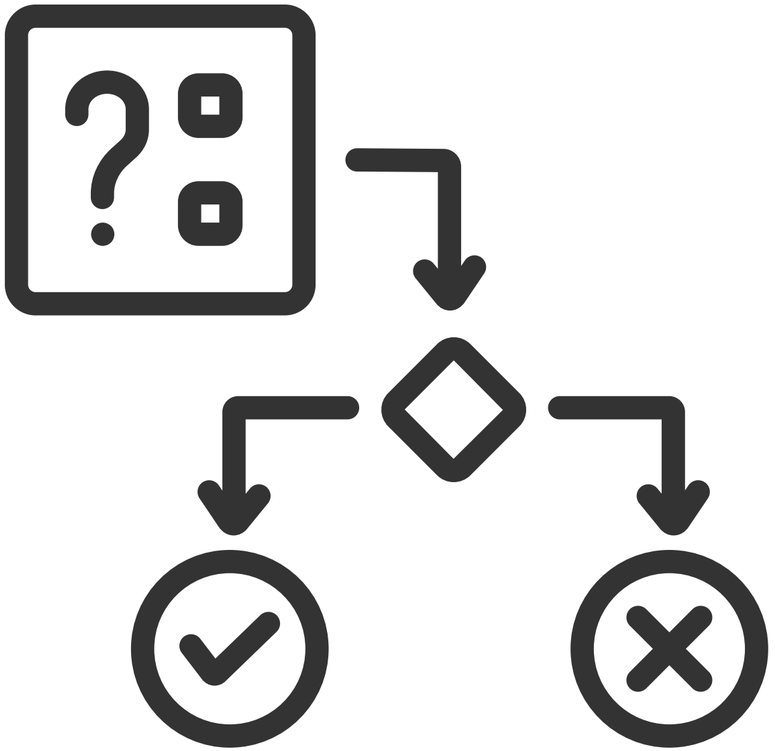
Effects

- Less collision
- Higher throughput
- Faster response
- Stable communication in motion

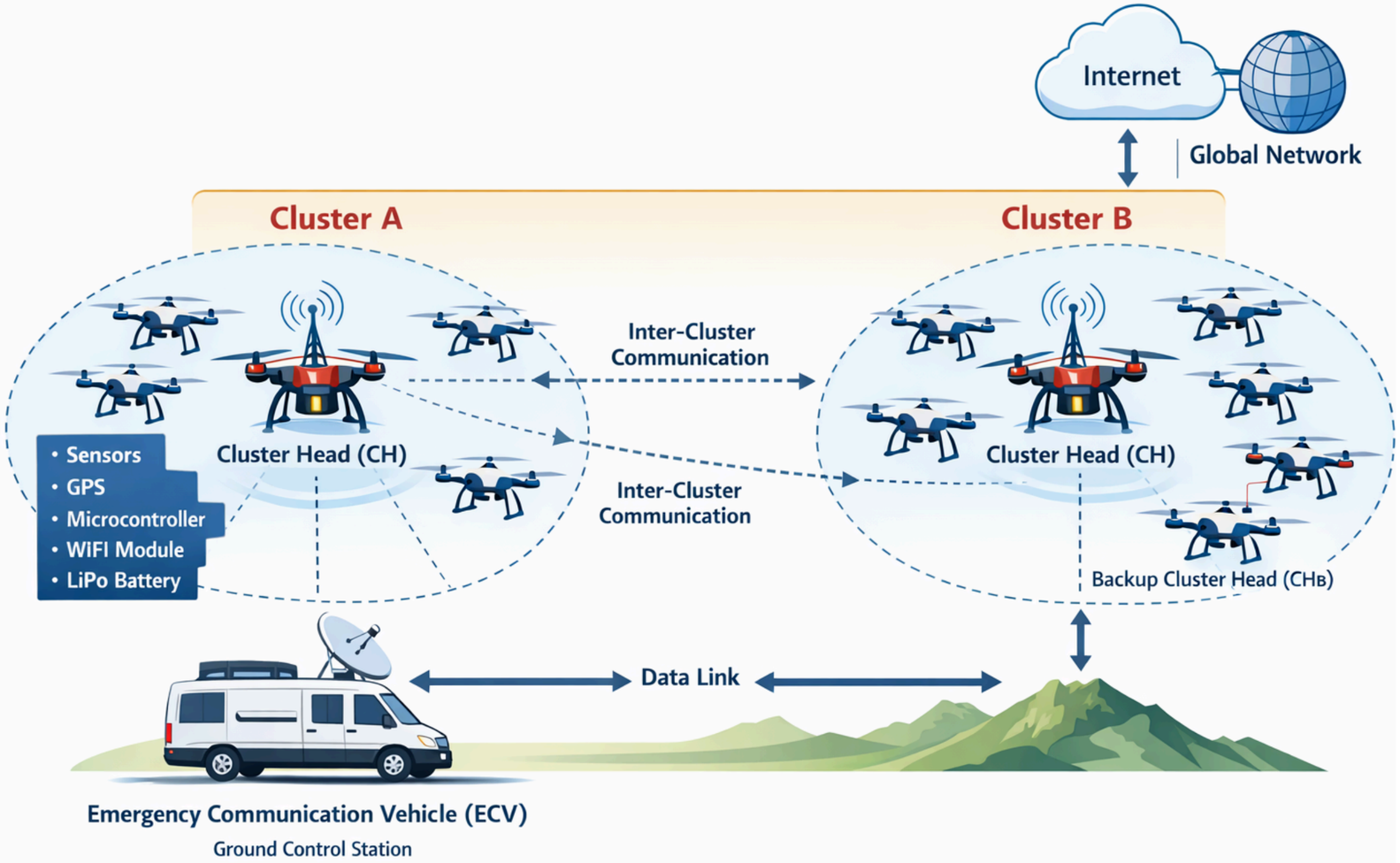


System Response Under Different Conditions

- If CH fails → CH_B takes control.
- If mobility increases → The network re-clusters.
- If congestion increases → Backoff window adjusted.



System Architecture



Three Main Layers

1. Emergency Communication Vehicle (Ground Control)
2. Cluster Heads (CH)
3. Cluster Members (CM – UAVs)

Block Diagram Overview

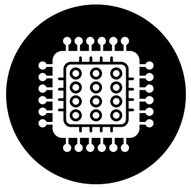
UAV Components



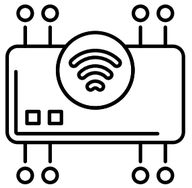
GPS Module



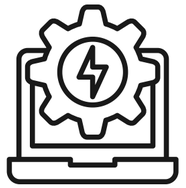
Environmental Sensors



Microcontroller



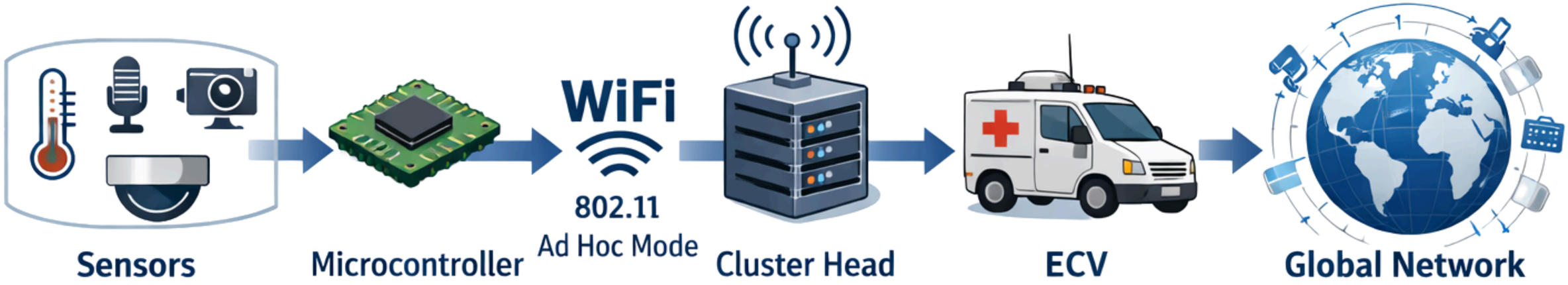
WiFi Module



Power System

Block Diagram Overview

Connection



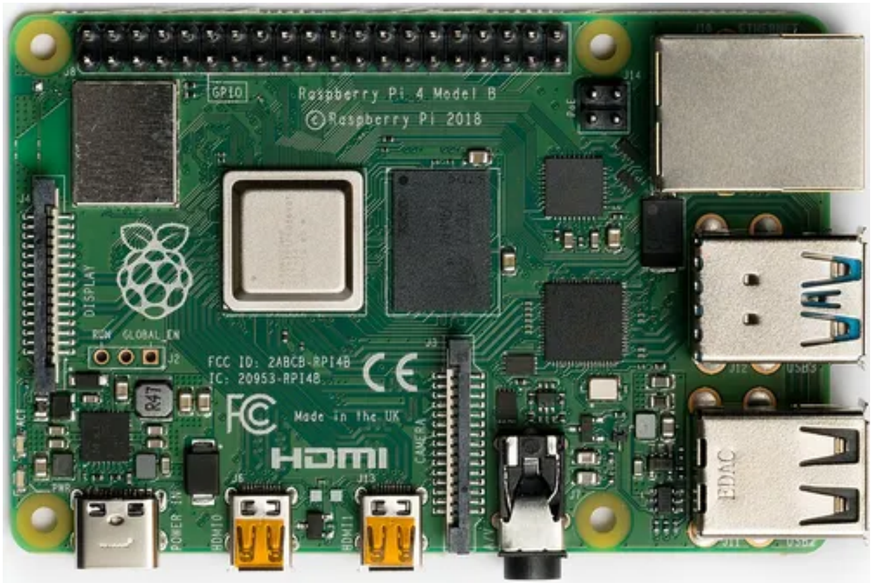
Communication Standard: IEEE 802.11 *(WiFi Ad Hoc Mode)*

Component Selection

Microcontroller

Raspberry Pi 4 / STM32MP1

Because of its **High Processing power** for cluster management & video relay



Alternative: ESP32

Component Selection

Communication

WiFi 6/6E

High data rate (11 Mbps), supports ad hoc



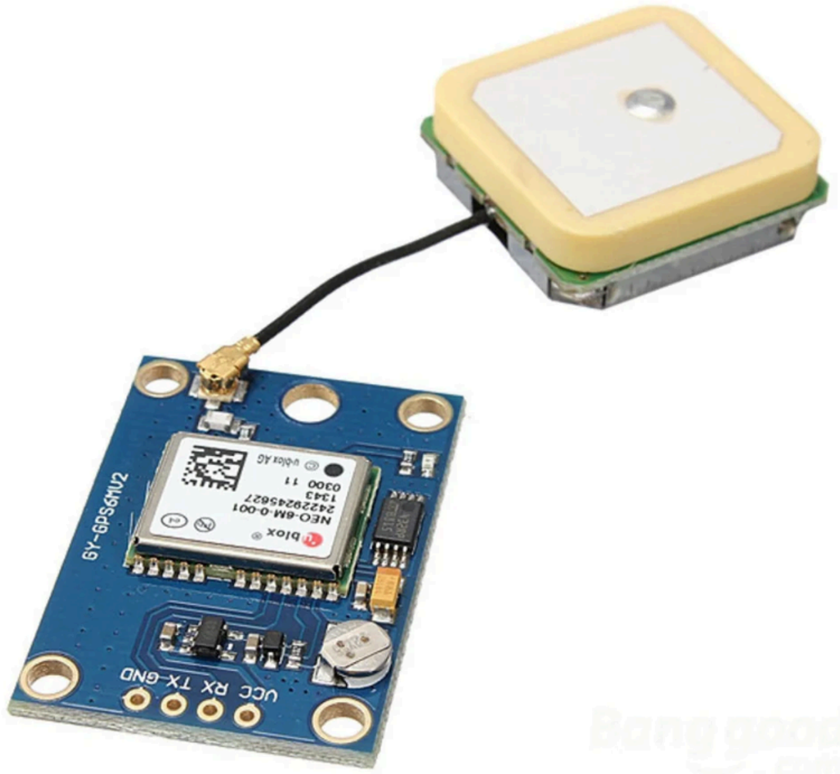
Alternative: LoRa (Too slow)

Component Selection

GPS

Neo-6M

Required for clustering & routing



Alternative: IMU

Component Selection

Sensors

IR, Temp, Pressure

Disaster monitoring



Alternative: Gas/Optical

Budget Analysis

Minimum Version Overview

Design

- Low-cost UAV
- ESP32 controller
- Standard WiFi

Removed Features

- No Backup CH
- Limited video streaming

Compromises

- Higher latency
- Lower throughput
- Single point of failure

Reliability

- If CH fails → Entire cluster fails

Budget Analysis

Best Quality Version Overview



Improvements

- Adaptive Backoff implemented
- Backup CH redundancy
- High-gain antennas
- Advanced GPS
- High-capacity battery



Performance

- Delay < 100ms guaranteed
- High throughput
- Improved reliability



Budget Impact

Increased cost due to:

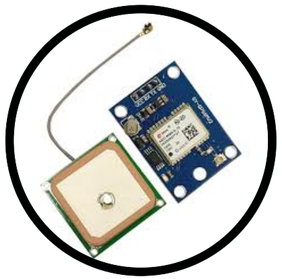
- Advanced controller
- Better antennas
- Extra redundancy hardware

Availability in Bangladesh

Components for the UAV system



Raspberry Pi 4



Neo-6M GPS



WiFi 6/6E



LiPo Battery

Local Sources

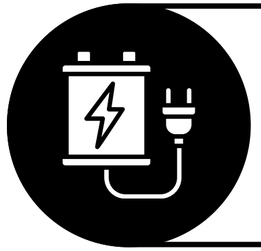
- Dhaka Patuatuly
- TechShopBD
- Stadium Market

Import Required

- Advanced 5G/6G modules
- High-gain long-range antennas

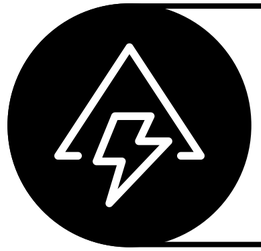
Power Supply Design

Essential battery system specifications



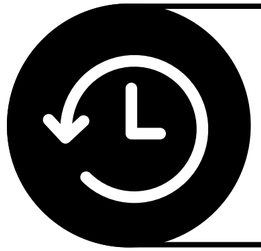
Power Source

3S/4S LiPo Battery



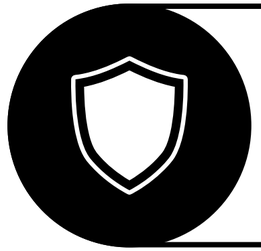
Voltage Requirements

5V → Controller | 12V → Motors



Estimated Runtime

20–30 minutes per UAV

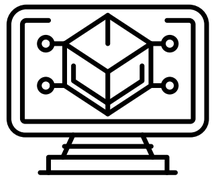


Protection

PDB | Voltage regulator | Fuse protection

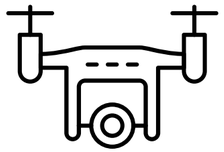
Development Plan

Steps for effective implementation



Simulation

- Use MATLAB
- Verify delay & throughput

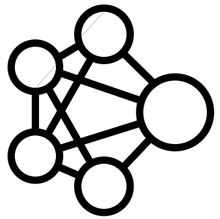


Prototype Setup

- Configure UAV with GPS & WiFi

Development Plan

Steps for effective implementation



Implement Clustering Logic

- RTJC / ATJC / RTBCH control packets



Field Testing

- Measure Packet Dropping Rate (PDR)
- Measure Safety Message Delay

System Limitations

- High Mobility → Frequent link breakage
- Limited battery life
- Signal Fading (Nakagami-m environment)
- Weather sensitivity

Future Improvements

- Integration of 6G communication
- AI-based trajectory optimization
- Autonomous obstacle avoidance
- Extended battery using solar-assisted UAV
- Edge computing for faster data processing

Thank You